

Maulana Azad National Institute of Technology, Bhopal

Department of Electrical Engineering

Class: B.Tech. IIV Sem.

Sub: Generation of Electrical Power (E-224)

Session- Jan-April. 2025

Assignments-4

1. A hydroelectric station is designed to operate at a mean head of 205 m and fed by a reservoir having a catchment area of 1000 km^2 with an annual rainfall of 125cm of which 80% is available for power generation. The expected load factor is 75%. Allowing a head loss of 5 m and assuming efficiency of turbine and generator to be 0.9 and 0.95 calculate suitable MW rating of the station. Comment on the type of turbine to be used.

2. A run-off data of a river is as under:

Month	Million m^3 per month	Month	Million m^3 per month
Jan.	80	July	150
Feb.	50	Aug.	200
March	40	Sep.	250
April	20	Oct.	120
May	0	Nov.	100
June	100	Dec.	80

- a) Plot hydrograph and flow duration curve. Find mean flow
b) Find power available at mean flow if head available in 100 m and efficiency of generation is 80 %. Take each month of 30 days.
3. A sample of uranium has 1.7×10^{24} atoms of U^{235} . If half life of U^{235} is 7.1×10^8 years, find decay constant, initial activity and number of U^{235} atoms remaining after 10×10^8 years.
4. Find the U^{235} fuel used in one year in a 235 MW pressurised water reactor. Assume an overall plant efficiency of 33 % and 100 % load factor throughout the year.

-----X-----

Maulana Azad National Institute of Technology, Bhopal

Department of Electrical Engineering

Class: B.Tech. IIV Sem.

Sub: Generation of Electrical Power (E-224)

Session- Jan-April. 2025

Assignments-5

1. A steam station and hydro station feed an area jointly. The hydro station is run for 14 hours daily and steam station is run for all the 24 hours. The production cost characteristics for steam station is

$$C = 5 + 8P_s + 0.05 P_s^2 \text{ Rs/hour}$$

If the load on the steam station, when both plants are in operation is 250 MW the incremental water rate of hydro plant is

$$\frac{dW}{dP_h} = 30 + 0.05 P_h \text{ m}^3/\text{MW-sec.}$$

The total quantity of water used during the 14 hours is 500 million cubic meters. Find the load on hydro plant and cost of water use. Assume that load on hydro plant is constant for the 14 hour period.

2. A two plant system having a steam plant near load centre and a hydro plant at a remote location is shown in Fig. The load is 700 MW for 14 hours a day and 500 MW for 10 hours a day .

The characteristics of units are

$$C_1 = (24 + 0.02 P_1) P_1 \text{ Rs/hour}$$

$$W_2 = (6 + 0.0025 P_2) P_2 \text{ Rs/hour}$$

Loss coefficient $B_{22} = 0.0005$.

Find the generation schedule, daily water used by hydro plant and daily operating cost of thermal plant for $r_2 = 2.5$ Rs. per hour/m³ per sec

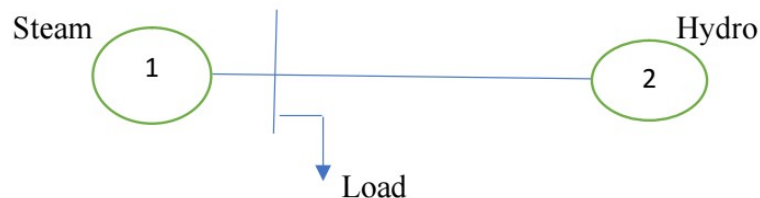


Fig. Hydro Thermal Systems

3. Two units of a thermal station have each the following cost characteristics

$$C = 5000 + 450P + 0.5P^2 \text{ Rs./hr}$$

Due to an instrumental error the cost characteristics of first unit is in error by +2% and that of the second unit by -2% at the time of scheduling. Find the extra operating cost due to erroneous scheduling. Total load is 200MW.

4. In a system consisting of two generating units connected through a transmission line, the incremental costs are :

$$\frac{dC_1}{dP_1} = 0.16P_1 + 32$$

in Rs/M-Wh

$$\frac{dC_2}{dP_2} = 0.24P_2 + 36$$

The system is operating on economic dispatch with $P_1 = P_2 = 100\text{MW}$ and

$$\frac{\partial P_L}{\partial P_2} = 0.2 \quad \text{Find the penalty factor of the plant.}$$

-----X-----